

Master's Exam: Theory Part. Answer all 4 questions. Show work in bluebook.

Saturday 6 December 2008

1. In a class of 30 students, 5 of the students play chess, and the other 25 do not. Four students are picked at random from the class. Given that at least one of the four students plays chess, what is the probability that at least two of the four students play chess?

2. Let X be a continuous random variable with probability density function $f(x) = 0.4x + 0.1$, for $0 \leq x \leq 2$. Let $Y = 3X - 4$.

a. Find $Var(Y)$.

b. Find the probability density function of Y .

3. Let Y have a binomial distribution with 10 trials and success probability π ; that is, $Y \sim B(10, \pi)$. Test $H : \pi = 1/2$ vs. $K : \pi > 1/2$. Suppose $Y = 9$. Find the p -value of a test for the null hypothesis H vs. the alternative hypothesis K . If the size of the test is $\alpha = .05$, do I accept or reject?

4. In a large national park, the number of raccoons in a randomly-picked acre is a Poisson random variable X , with $E(X) = \lambda$. Given x raccoons in a particular acre, the “number of raccoons that have rabies” is Y , a binomial random variable:

$$P(Y = y|X = x) = \frac{x!}{y!(x-y)!} \pi^y (1-\pi)^{x-y}.$$

We draw a random acre and find that there are exactly 4 raccoons with rabies. Given this, and given $\lambda = 5$, $\pi = .3$, what is the expected number of raccoons on the acre.